178067: recrystallized sandstone, Four Corners Bore

Location and sampling

YOUANMI (SH 50-4), EVERETT CREEK (2841)
MGA Zone 50, 769640E 6861250N

Sampled on 31 October 2002

The sample was taken from a 0.5 m-diameter block on the eastern side and near the top of a north-northwesterly trending rocky ridge, 4 km south of Four Corners Bore. The sampling site is 100 m east of the site of GSWA sample 178068 (dated by Nelson, this compilation).

Tectonic unit/relations

The sample is a massive, greyish-white, coarse- and even-grained, recrystallized quartz sandstone, with a possible quartz vein up to 1 mm thick, from a bedded quartzite unit in the Maynard Hills greenstone belt of the Southern Cross Granite–Greenstone Terrane, Yilgarn Craton. There is no established stratigraphy for this belt (Riganti, 2003). Bedding in the quartzite unit strikes at 330° and dips at 50° to the west. The sample was collected to constrain the maximum depositional age of the sandstone precursor to the quartzite.

Petrographic description

The principal minerals are quartz (98 vol.%) and fine schistose mica (≤1 vol.%), with accessory rutile (trace) and partly clouded, fractured zircon (trace). The sample consists of a metamorphic quartz mosaic with interlocking anhedral grains about 10 to 15 mm long, and traces of very fine grained schistose mica, mostly less than 0.1 mm in crystal size. Rare rutile and accessory zircon are up to 70 µm in diameter. Slightly larger ?mica flakes have been leached out, and larger ?zircon grains are fractured, clouded, and altered. Fluid inclusions in the quartz appear to be rich in CO_2.

Zircon morphology

The zircons isolated from this sample are typically colourless, pale yellow, dark brown or black, elliptical or subhedral and with rounded terminations, equant or slightly to substantially elongate, and between 50 × 80 µm and 160 × 80 µm in size. Most grains are fractured and internally structureless, but some have faint internal zonation. Many grains have a speckled appearance in transmitted light, and many are metamict. The surfaces of many grains are pitted, consistent with detrital transport. In reflected light, many surfaces have irregular, dark patches and zones. Cathodoluminescence images of representative zircons are given in Figure 1.

Analytical details

This sample was analysed on 20 June and 20 October 2003. The counter deadtime during both analysis sessions was 24 ns. Fourteen analyses of the CZ3 standard obtained during the first analysis session indicated a Pb*/U calibration uncertainty of 1.39% (1σ). Analyses 1.1 to 15.1 were obtained during the first analysis session. Seven analyses of the CZ3 standard obtained during the second analysis session indicated a Pb*/U calibration uncertainty of 0.918% (1σ). A calibration uncertainty of 1.0% (1σ) was applied to analyses of unknowns obtained during the second analysis session. Common-Pb corrections were applied assuming Broken Hill common-Pb isotopic compositions for all analyses.

Results

Thirty-nine analyses were obtained from 36 zircons. Results are given in Table 1, and shown on concordia and Gaussian-summation probability density plots in Figures 2 and 3 respectively.

Interpretation

The analyses are concordant to highly discordant, with the discordance pattern consistent with a recent episode of radiogenic-Pb loss. On the basis of their $^{207}Pb/^{206}Pb$ ratios, many analyses can be assigned to one of five groups. Concordant analysis 11.1 and discordant analyses 5.1, 24.1, 24.2, 24.3, and 35.1, assigned to Group 1, have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 3283 ± 11 Ma (chi-squared = 1.43). Concordant and near-concordant analyses 6.1, 9.1, 12.1, 17.1, 20.1, and 26.1, assigned to Group 2, have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 3325 ± 10 Ma (chi-squared = 0.55). Thirteen concordant to highly discordant analyses of 13 zircons (1.1, 3.1, 8.1, 13.1, 15.1, 16.1,19.1, 23.1, 25.1, 27.1, 28.1, 30.1, 36.1), assigned to Group 3, have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 3383 ± 7 Ma (chi-squared = 1.37). Concordant analysis 22.1, and discordant analyses 31.1 and 32.1, assigned to Group 4, have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 3413 ± 23 Ma (chi-squared = 0.17). Near-concordant analyses 7.1, 18.1, 29.1, and 33.1, assigned to Group 5, have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 3693 ± 13 Ma (chi-squared = 0.38). The remaining analyses cannot be confidently grouped.

The analyses are interpreted to be of detrital zircons.

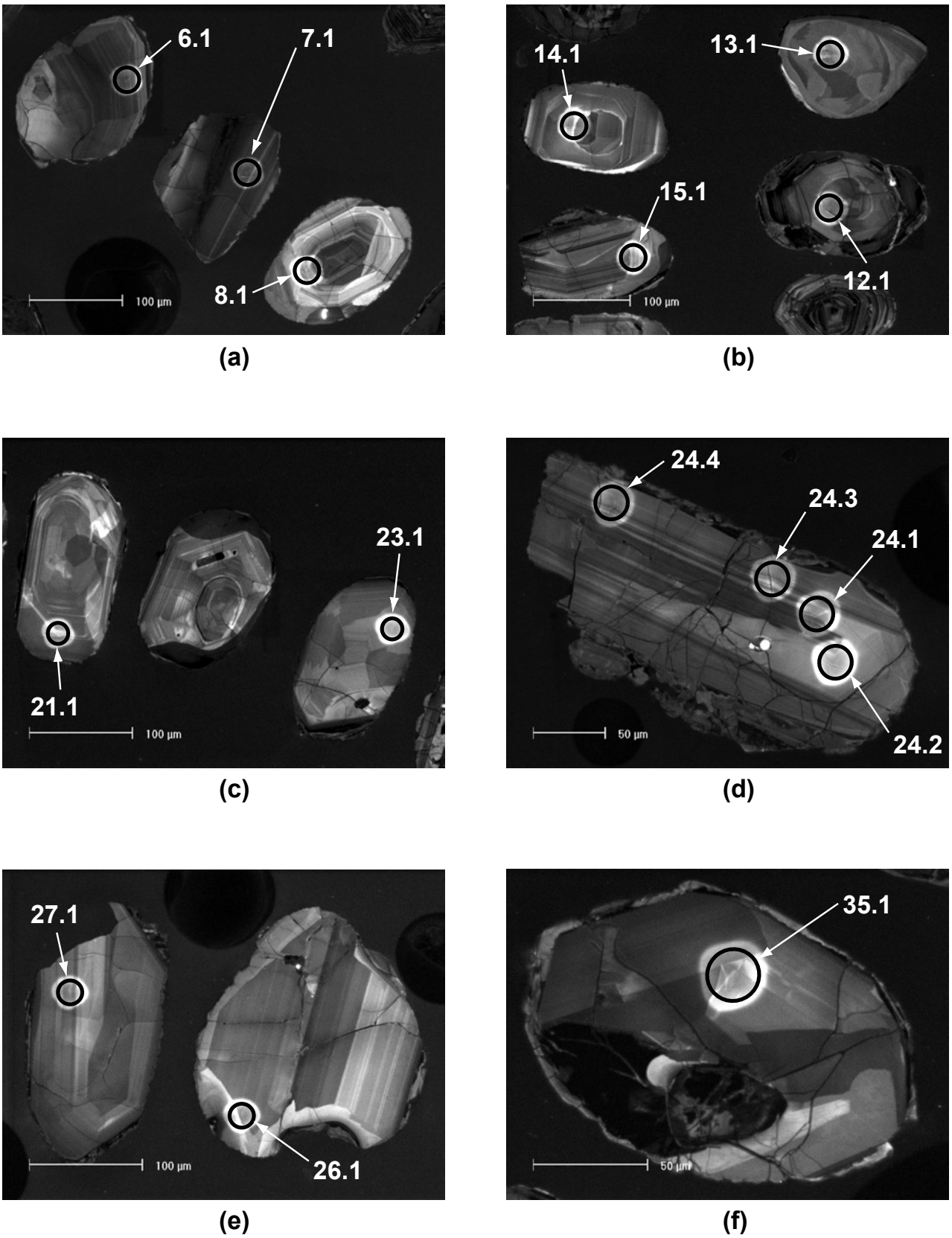

Figure 1. **Cathodoluminescence images of representative zircons from sample 178067: recrystallized sandstone, Four Corners Bore**

Table 1. Ion microprobe analytical results for zircons from sample 178067: recrystallized sandstone, Four Corners Bore

Grain .spot	*U (ppm)*	*Th (ppm)*	*Pb (ppm)*	*f206%*	$^{207}Pb/^{206}Pb$	±1σ	$^{208}Pb/^{206}Pb$	±1σ	$^{206}Pb/^{238}U$	±1σ	$^{207}Pb/^{235}U$	±1σ	*% concordance*	$^{207}Pb/^{206}Pb$ *Age*	±1σ
1.1	55	32	47	-0.060	0.28182	0.00213	0.15289	0.00258	0.6955	0.0115	27.025	0.513	101	3 373	12
2.1	71	70	75	0.090	0.38816	0.00238	0.23761	0.00323	0.7529	0.0120	40.292	0.716	94	3 864	9
3.1	140	151	132	0.128	0.28613	0.00136	0.28340	0.00192	0.6907	0.0103	27.248	0.441	100	3 397	7
4.1	101	25	95	0.115	0.34057	0.00171	0.06292	0.00153	0.7739	0.0118	36.342	0.605	101	3 665	8
5.1	293	263	73	0.305	0.26557	0.00157	0.11997	0.00196	0.2063	0.0029	7.552	0.121	37	3 280	9
6.1	137	34	109	-0.038	0.27543	0.00205	0.06949	0.00373	0.6886	0.0103	26.150	0.460	101	3 337	12
7.1	130	199	153	0.059	0.34686	0.00143	0.38976	0.00203	0.7831	0.0118	37.452	0.601	101	3 693	6
8.1	53	39	47	0.219	0.28037	0.00252	0.19181	0.00406	0.7019	0.0117	27.133	0.538	102	3 365	14
9.1	109	134	103	0.135	0.27458	0.00146	0.32395	0.00222	0.6827	0.0104	25.847	0.433	101	3 332	8
10.1	12	8	11	-0.591	0.31937	0.00661	0.20566	0.01163	0.7449	0.0180	32.803	1.104	101	3 567	32
11.1	183	66	142	0.057	0.26717	0.00113	0.09531	0.00110	0.6623	0.0097	24.398	0.385	100	3 290	7
12.1	138	151	125	0.078	0.27196	0.00134	0.28567	0.00200	0.6783	0.0101	25.434	0.414	101	3 317	8
13.1	96	44	80	-0.014	0.28393	0.00161	0.12462	0.00169	0.6905	0.0106	27.031	0.461	100	3 385	9
14.1	82	57	74	0.110	0.30473	0.00192	0.18157	0.00258	0.7039	0.0111	29.577	0.521	98	3 494	10
15.1	90	41	76	0.104	0.28107	0.00166	0.11791	0.00181	0.6942	0.0107	26.904	0.463	101	3 369	9
16.1	121	109	107	0.280	0.28362	0.00199	0.23870	0.00293	0.6737	0.0083	26.344	0.394	98	3 383	11
17.1	215	264	175	0.138	0.27352	0.00149	0.27981	0.00228	0.6068	0.0069	22.884	0.302	92	3 326	9
18.1	98	116	107	0.203	0.34787	0.00251	0.31430	0.00387	0.7609	0.0098	36.496	0.567	99	3 698	11
19.1	145	48	115	0.235	0.28287	0.00179	0.08596	0.00206	0.6700	0.0081	26.133	0.373	98	3 379	10
20.1	151	138	96	0.293	0.27288	0.00198	0.20504	0.00290	0.4921	0.0058	18.514	0.271	78	3 323	11
21.1	76	58	71	0.333	0.30777	0.00266	0.20443	0.00374	0.7046	0.0096	29.902	0.508	98	3 510	13
22.1	163	133	148	0.336	0.28963	0.00177	0.21100	0.00255	0.6984	0.0082	27.889	0.389	100	3 416	10
23.1	72	70	65	0.546	0.28433	0.00295	0.26261	0.00510	0.6701	0.0092	26.271	0.479	98	3 387	16
24.1	75	105	68	0.392	0.26422	0.00304	0.36632	0.00588	0.6314	0.0084	23.002	0.430	96	3 272	18
25.1	71	57	64	0.762	0.28034	0.00288	0.20493	0.00466	0.6928	0.0096	26.780	0.488	101	3 365	16
26.1	82	87	73	0.284	0.27194	0.00237	0.27898	0.00372	0.6616	0.0088	24.807	0.416	99	3 317	14
27.1	134	117	119	0.371	0.28409	0.00190	0.22938	0.00276	0.6731	0.0081	26.365	0.381	98	3 386	10
28.1	289	315	169	0.240	0.28217	0.00143	0.17759	0.00188	0.4611	0.0051	17.941	0.227	72	3 375	8
29.1	188	75	180	0.330	0.34527	0.00170	0.09779	0.00199	0.7600	0.0088	36.180	0.475	99	3 686	8
30.1	63	60	57	0.610	0.29099	0.00303	0.26209	0.00493	0.6695	0.0096	26.862	0.504	97	3 423	16
31.1	113	50	94	0.280	0.28990	0.00202	0.11608	0.00238	0.6809	0.0084	27.215	0.407	98	3 417	11
32.1	219	235	202	0.240	0.28843	0.00145	0.28245	0.00217	0.6765	0.0077	26.902	0.349	98	3 409	8
33.1	109	147	120	0.102	0.34818	0.00212	0.35276	0.00296	0.7481	0.0094	35.915	0.525	97	3 699	9
34.1	295	138	287	0.170	0.35264	0.00129	0.12025	0.00120	0.7634	0.0084	37.118	0.447	98	3 718	6
35.1	120	65	95	0.466	0.26813	0.00207	0.14123	0.00293	0.6416	0.0079	23.719	0.363	97	3 295	12
36.1	241	199	215	0.160	0.28480	0.00132	0.21788	0.00172	0.6864	0.0077	26.953	0.342	99	3 389	7
24.2	40	38	34	0.537	0.26539	0.00350	0.26062	0.00667	0.6483	0.0089	23.724	0.479	98	3 279	21
24.3	105	155	75	0.425	0.26307	0.00175	0.34720	0.00305	0.5142	0.0059	18.650	0.260	82	3 265	10
24.4	126	272	69	0.464	0.25255	0.00176	0.34780	0.00320	0.3945	0.0043	13.738	0.189	67	3 201	11

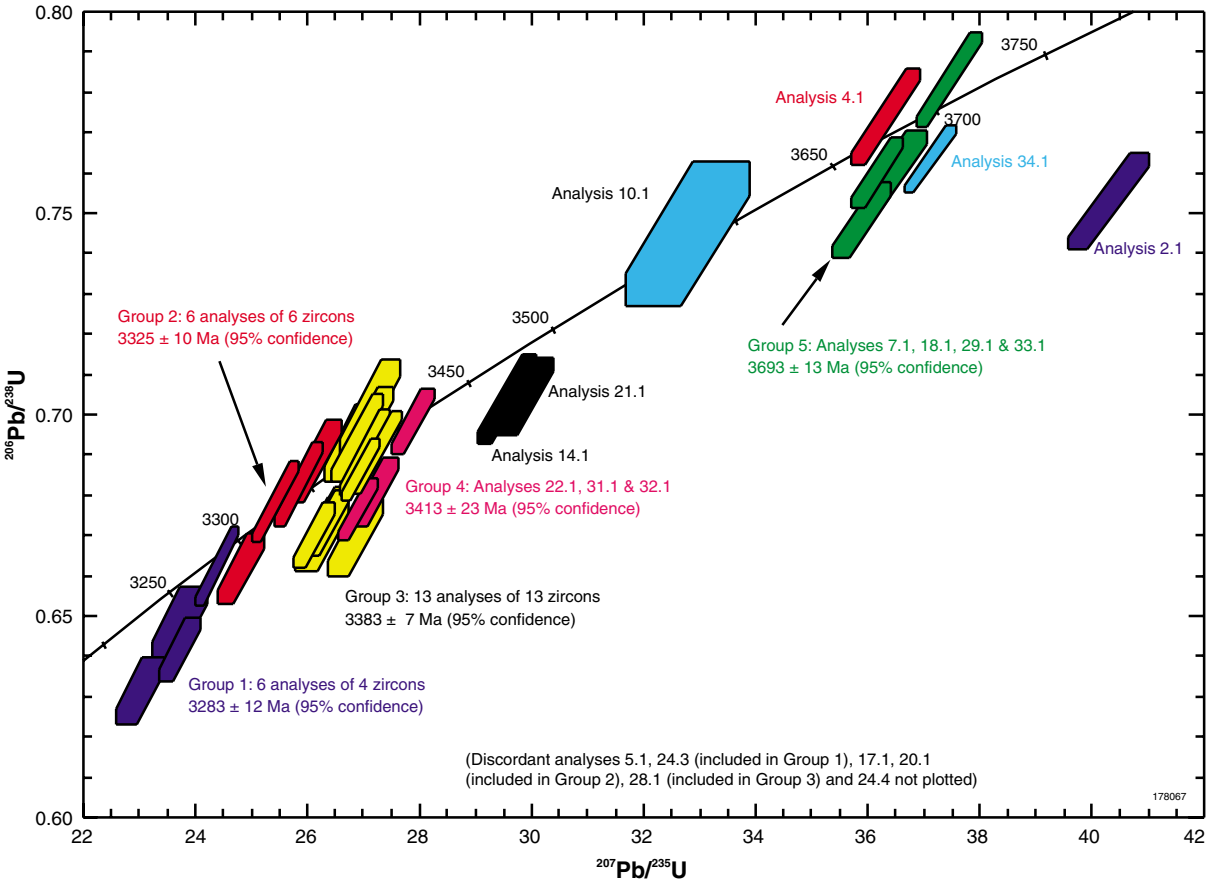

Figure 2. Concordia plot for sample 178067: recrystallized sandstone, Four Corners Bore

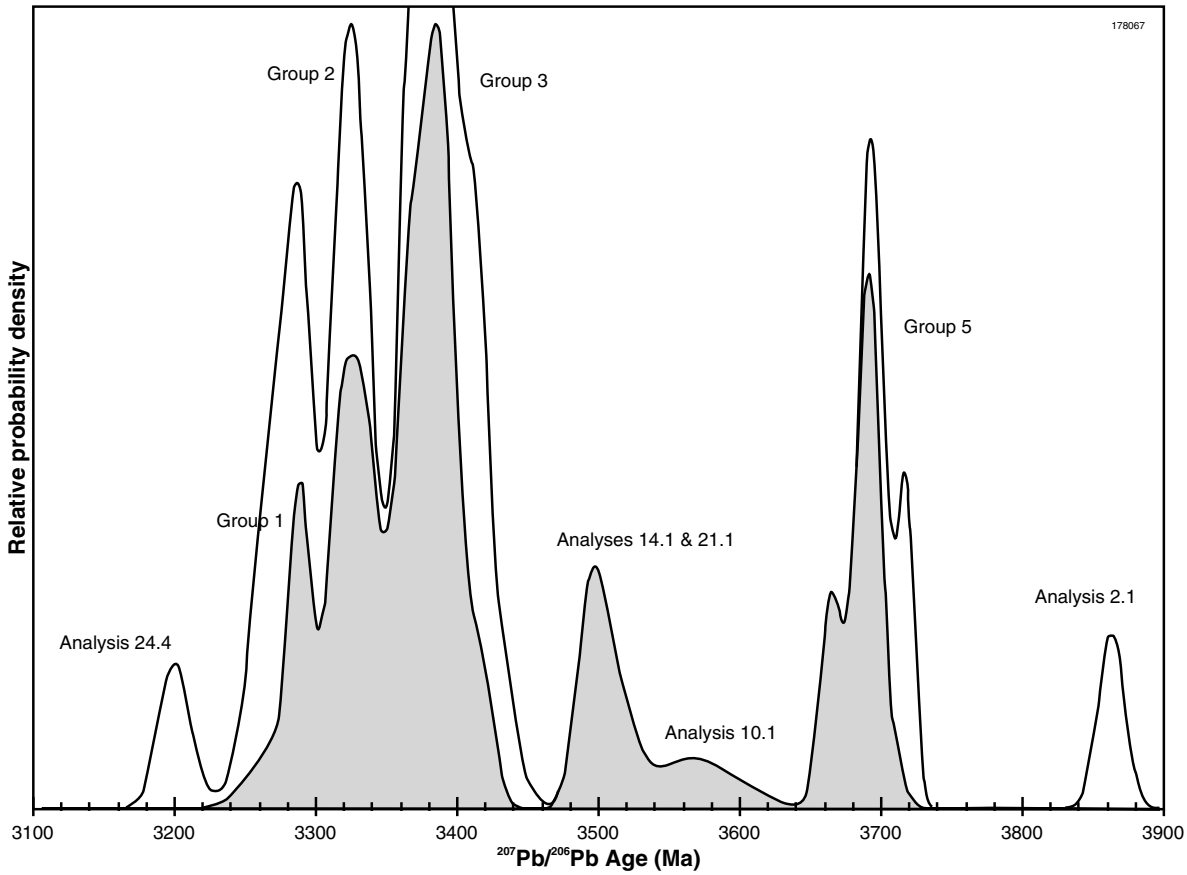

Figure 3. Gaussian-summation probability density plot for sample 178067: recrystallized sandstone, Four Corners Bore

The date of 3283 ± 11 Ma indicated by the weighted mean $^{207}Pb/^{206}Pb$ ratio of concordant analysis 11.1 and discordant analyses 5.1, 24.1, 24.2, 24.3, and 35.1 of Group 1 is therefore interpreted as providing a maximum age for deposition of the precursor sandstone. Source rocks within the western part of Australia having ages matching those of the zircons within this sample include the Narryer Terrane in the northwestern part of the Yilgarn Craton (Nelson, 1995, 1996, 1997) and the eastern part of the Pilbara Craton (Nelson, 1996, 1997, 1998, 1999, 2000, 2001, 2002, this compilation).

Recommended reference for this publication:

NELSON, D. R., 2005, 178067: recrystallized sandstone, Four Corners Bore; Geochronology dataset 594; in Compilation of geochronology data, June 2006 update: Western Australia Geological Survey.

Data obtained: 20/10/2003; Data released: 30/06/2005